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Proning face pack

Abstract

A face support pack for controlling moisture and friction induced skin breakdown that includes a moisture vapor removal layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/176,188 titled “Support Packs and Methods of Making the Same” filed Apr. 16, 2021. The full disclosure of the aforementioned patent application is herein fully incorporated by reference FIELD (2) This invention relates to patient face support for prone therapy.

BACKGROUND

(1) There is a need for an improved face support pack useful for controlling moisture and friction-induced skin breakdown during prone therapy.

SUMMARY

(2) A proning head pack comprising a facial interface layer configured to support a patient face, the facial interface layer being liquid vapor permeable but liquid and air impermeable; a boundary layer being air impermeable; and a spacer layer disposed between said interface layer and said boundary layer, the spacer layer configured to air flow therethrough; the facial interface layer and boundary layer each having a perimeter shaped so as to leave exposed one or more face orifices when the facial interface layer is disposed against the patient face, the facial interface layer and boundary layer being joined along the perimeter so as to seal the spacer layer between the facial interface layer and the boundary layer.

(3) A proning head pack configured to leave exposed the eyes, mouth and nose of the patient face when disposed against a patient face, the head pack comprising a moisture vapor removal layer configured for interface with the patient face, the moisture vapor removal layer configured to transmit moisture vapor away from the patient face but not transmit air or liquid; and a support layer.

(4) A proning head pack configured to leave exposed the eyes, mouth and nose of the patient face when disposed against a patient face, the head pack comprising a moisture vapor removal layer configured for interface with the patient face, the moisture vapor removal layer configured to transmit moisture vapor away from the patient face but not transmit air or liquid, the moisture vapor removal layer being further configured to receive fluid pressure from a fluid pressure source so as to provide support to the patient when using the face pack; a vent disposed so as to vent moisture vapor from the moisture vapor removal layer to atmosphere; and a valve configured to pulse open so as to provide cyclical variation of pressure within the moisture vapor removal layer.

(5) A proning head pack comprising a facial interface layer configured to support a patient face, the facial interface layer being liquid vapor permeable but liquid and air impermeable; a boundary layer being air impermeable and substantially impermeable to water vapor; and a spacer layer disposed between said interface layer and said boundary layer, the spacer layer configured to air flow therethrough; wherein said boundary layer includes one or more openings allowing for transport of air out of said boundary layer.

(6) A proning head pack configured to leave exposed the eyes, mouth and nose of the patient face when disposed against a patient face, the head pack comprising a moisture vapor removal layer configured for interface with the patient face, the moisture vapor removal layer configured to transmit moisture vapor away from the patient face but not transmit air or liquid, the layer being further configured to receive fluid pressure from a fluid pressure source so as to provide support to the patient when using the face pack, the moisture vapor removal layer including a facial interface layer, the facial interface layer being liquid vapor permeable but liquid and air impermeable; a boundary layer being air impermeable, the boundary layer including one or more openings to provide for selective communication of moisture vapor from the moisture vapor removal layer to a support layer; and a spacer layer disposed between the interface layer and the boundary layer, the spacer layer configured to air flow therethrough; said support layer being configured to help support the patient even if pressure is lost from the head pack.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates an embodiment of a therapeutic bed configured for prone therapy.
(2) FIG. **2** illustrates an embodiment of a head pack support assembly shown with a patient in a supine or face up position.
(3) FIG. **3** illustrates a head pack support assembly shown with a patient in a prone or
(4) face down position.
(5) FIG. **4** illustrates an embodiment of a face pack.
(6) FIG. **5** illustrates moisture vapor removal.
(7) FIG. **6** illustrates another embodiment of a face pack wherein air flow is routed through a support layer.
(8) FIG. **7** illustrates assembly of an embodiment of a moisture vapor removal layer.
(9) FIG. **8** illustrates a method of assembling an embodiment of a moisture vapor removal layer.
(10) FIG. **9** illustrates another embodiment of a face pack including a check valve.
(11) FIG. **10** illustrates another embodiment of a face pack including a valve.
(12) FIG. **11** illustrates another embodiment of a face pack.
(13) FIG. **12** illustrates still another embodiment of a face pack.
DETAILED DESCRIPTION
(14) This disclosure is directed to face support packs for therapeutic beds configured for prone therapy. To provide context for describing the structure and function of various embodiments of face support packs, the disclosure turns first to an overview of an embodiment of a therapeutic bed in which a face support pack may be suitably provided.
(15) Therapeutic Bed
(16) FIG. **1** illustrates an embodiment of a therapeutic bed **10** configured to support a patient (not shown) for prone therapy and/or kinetic therapy. Therapeutic bed includes a patient support frame **12** having a head end **14** and a foot end **16**. The patient support frame is coupled to a caster frame **18** by a first lift column **20** at the head end and by a second lift column **22** at the foot end. The caster frame may be supported by a plurality of casters **24** for bed mobility.
(17) The therapeutic bed embodiment of FIG. **1** may move a patient through primarily two therapeutic modes of movement: a rotational mode and a tilt mode. To provide a rotational mode of movement, the patient support frame may be rotated about a long axis extending through the foot end and the head end of the patient support frame. The rotational mode of movement permits a patient to be rotated from a supine (face up) orientation to a prone (face down) orientation. The rotational mode of movement may further permit a patient to be oscillated through a range of angular positions in either or both of the supine or prone orientations. The rotational mode of movement may permit 360° rotation or more or less than 360° rotation.
(18) To permit rotational movement, the patient support frame may be rotatably coupled to the lift columns. For example, the foot end of the patient support frame may be coupled to lift column **22** by any suitable means, such as through a plate or saddle (not shown). Other suitable means for providing rotatable coupling between the lift column **22** and patient support frame may be used, such as those described in U.S. Pat. No. 6,862,759, for example, which is herein incorporated by reference. The head end of the patient support frame may comprise a hoop **25**, which may be coupled to a lift column **20** using any suitable means. For example, the patient support frame may rest on a roller support coupled to a saddle (not shown) with the saddle coupled to the lift column **20**. A drive system (not shown), such as an electrical motor and drive belt, and electronic controls may be used to selectively rotate the patient support frame. Of course, other suitable means for rotatably coupling the patient support frame and lift column **20** may be used. In some modes of operation, the patient support frame may be manually rotated.
(19) To provide a tilt mode of movement, the length of each lift column may be independently adjusted so as to raise and lower the head end of the patient support frame independently of the foot

end, or to raise and lower the foot end of the patient support frame independently of the head end. Furthermore, the length of each lift column may be adjusted so as to raise or lower the entire patient support frame with respect to the caster frame. That is, the distance between either or both end of the patient support frame and the caster frame may be adjusted. To permit tilt movement, lift column height may be adjusted by any suitable mechanism, such as by hydraulics, screw, gas spring, coil spring, ratchet or removable pin.
(20) Patient Constraint
(21) When the patient support frame is oriented to support a patient in a supine position, the patient may rest on one or more patient support pads **23** disposed on the patient support frame **12**. The one or more support pads **23** may provide a patient support surface **26** to support the patient (not shown in FIG. **1**). However, when the patient support frame **12** is moved through one or more modes of movement, the patient must be constrained from sliding or falling from the patient support frame. A variety of packs may be provided to constrain a patient during bed movement.
(22) A plurality of lateral packs may constrain the patient's legs, torso, arms and head from lateral movement with respect to the patient support surface. Such lateral packs may include, for example, side support packs **28**, foot packs **30**, abductor packs **32**, and head packs **36**.
(23) A plurality of prone packs may prevent a patient from falling from the bed when the patient is rotated to a prone position. Such prone packs may include, for example, leg packs **38**, torso or thigh packs **40** and a face pack **42**. Various embodiments of a head pack are described in more detail below.
(24) The term “pack” as used herein refers to a structure that is firm enough to substantially maintain its shape while supporting the patient's body but is also soft so as to comfortably support the patient's body. A pack may, for example, be comprised of a rigid support panel or other structure surrounded by a padding. A pack may be comprised of one or more layers. A pack may comprise a single type of padding. Alternatively, a pack may comprise several different padding materials such as may be used such as to provide a desired level of support in different parts of a pack. For example, a pack may be comprised of materials with more than one spring rate or initial force deflection rating so as to control a level of immersion of the pack around the patient's body. A pack may be shaped to receive a part of the patient's body. For example, a support pack may be generally shaped to contour a patient's legs, forehead, cheeks, or other body part against which it is designed to be disposed. In some embodiments, a pack may be shaped and/or made of materials with controlled properties (e.g., initial force deflection, spring rate, and other properties) so as to reduce any shearing stresses that tend to be formed on the patient's skin when a patient's body is immersed in the pack. A pack may, for example, be filled with a pressurized gas (such as air), foam, a gel, a viscous fluid, or another suitable material.
(25) Patient Access
(26) When the patient support frame is rotated to orient a patient in the prone position, a caregiver may require access to the patient through the patient support frame. The patient support frame may be provided with panels that a caregiver may open to allow access to the patient's body.
(27) Head Support Packs
(28) In view of the foregoing context, a more detailed description of various embodiments of head packs may now be provided. However, the foregoing embodiments of therapeutic beds and various features and functions thereof should not be interpreted as limiting. Any head pack as described herein may be used with any therapeutic bed in which a patient may be positioned or placed in a prone or face down position or in which a patient may be otherwise treated with rotation therapy.
(29) As may be seen in the embodiment of FIG. **2**, a head pack may designed for supporting the face side or the ear side or posterior of a patient's head. Thus, the head pack may also be referred to as a posterior head pack, a lateral or side head pack, or as a face pack, depending on the portion of the head for which the head pack is configured. As shown in FIG. **2**, the face pack **42** may be provided as part of a head pack support assembly **100**. The head pack support assembly **100** may

comprise a chassis **102** upon which various structures (e.g., securing straps and various packs) may be mounted for releasably constraining the head **104** of a patient **106** during therapeutic treatment of the patient. In the embodiment shown in FIG. **2**, chassis **102** may support the face pack **42**, one or more lateral head packs **36**, and a posterior head pack **108**. The face pack **42** may, for example, be releasably coupled to the chassis **102** by a shroud **112** retained by a plurality of straps **110**.

(30) The face pack **42** may be shaped to conform to the patient's face while leaving free the patient's eyes, nose and mouth. Thus, when the patient is in a prone position, as may be seen in the embodiment of FIG. **3**, the patient may be free to breathe and see through the face pack **42**. While novel aspects of MVTR (moisture vapor transmission rate) packs are described in an exemplary manner with respect to the face pack **42**, it should be understood some embodiments herein may be applied with other head packs, such as the one or more lateral head packs **36** and posterior head pack **108**. Furthermore, in some situations, patients may be oriented on there side during therapeutic treatment. In those situations, a lateral pack may sometimes include car cutouts so that a weight of a patient's head when oriented on a side is supported accordingly without having pressure applied to the patient's ears. The face pack **42** may be shaped conform to the width and height of the patient's face such that the outer dimensions of the face pack **42** approximately fit the patient's face. As may be seen in the embodiment of FIG. **2**, the face pack **42** may have a width across the patient's face to extend to the side of the patient's face or just past the ears of the patient so as to permit coupling of the face pack **42** to the chassis **102** by the plurality of straps **110**. The face pack **42** may be of sufficient width to permit the face pack to be secured against lateral head packs **36**. As may be seen in the embodiment of FIG. **2**, the face pack **42** may have a height along the patient's face to extend to or just past the patient's chin and extend to cover the patient's forehead. The face pack may thus have a perimeter shaped so as to leave exposed the eyes, nose and mouth of the patient and to conform to the outer dimensions of the patient's face when the face pack is disposed against the patient face. By configuring the face pack to maximize interface with a patient's face yet leave an opening for the eyes, nose and mouth of the patient, interface pressure may be spread over a larger surface area and thereby reduced.

(31) As may be seen in FIG. **4**, the face pack **42** may comprise a layer **120** configured to transfer moisture vapor away from a patient's face at a rate (such rate generally being referred to as a moisture vapor transfer rate or MVTR) suitable for reducing breakdown of facial skin due to skin-to-pack friction and moisture trapped between the face and head pack (such layer being referred to herein as an MVR layer or a MVTR layer) and a support layer **122**. Air may flow into the MVTR layer **120** through an inlet **114** to permit moisture vapor removal through the interface between the patient's face and the face pack. Air may be provided by an air mover **116**, such as a blower or pump, and connecting hose **118**. Air and moisture vapor may flow out of the face pack **42** to atmosphere through a vent (not shown) or air and moisture or water vapor may be removed by other means.

(32) Moisture Vapor Removal Layer

(33) As may be seen in the embodiment of FIG. **5**, the MVTR layer **120** may comprise a facial interface layer **130**, a spacer layer **132** and a boundary layer **134**. The facial interface layer **130** may be permeable to moisture vapor, but substantially impermeable to air and most liquids. The spacer layer **132** comprises an air-permeable material, such as a low-density foam, used to direct air flow as appropriate for routing moisture vapor in a controlled manner away from the patient's face. For example, the spacer layer **132** may route air flow from an inlet to a vent **143** so that moisture-vapor is driven out of the spacer layer **132** through the vent **143**. Removal of water vapor **142** from the spacer layer **132** will help to encourage further net transport of moisture across the facial interface layer **130**. The boundary layer **134** may be substantially impermeable to air so as to encourage proper air flow for removing moisture from the spacer layer **132**. In some embodiments, it may be desirable to route moisture for removal without allowing significant amounts of moisture vapor to infiltrate an underlying support layer **122**. In those embodiments, the boundary layer **134** may be

substantially impermeable to water vapor. However, in other embodiments, the boundary layer **134** may be permeable to moisture vapor.

(34) In some embodiments, the MVTR layer **120** may operate as shown in FIG. **5**. The MVTR layer **120** may be disposed on a support layer **122**. A patient's face (not shown) may generate moisture. The facial interface layer **130** may prevent liquid moisture **138** from entering the MVTR layer **120** but will allow moisture vapor **142** to pass into the MVTR layer. Air **140** forced through the MVTR layer **120** will mix with moisture vapor **142** as it passes through the MVTR layer. The air **140** and moisture vapor **142** (i.e., moisture-saturated or vapor enriched air), will exit the MVTR layer **120** and direct moisture away from the patient such as to atmosphere or another moisture sink.

(35) In some embodiments, moisture-saturated or moisture enriched air in the MVTR layer **120** may be exhausted directly into the support layer **122** or pass through the support layer via one or more passages formed therein so that moisture vapor is removed from the spacer layer **132**. As opposed to removing moisture vapor through a passage formed in the support layer **122**, the support layer may be made with one or more holes or openings formed therein. Moisture vapor **142** may then be actively driven or passively moved through the openings and vented to atmosphere or into another moisture vapor sink.

(36) In some embodiments, the boundary layer **134** will prevent air **140** and moisture vapor **142** from passing through to the support layer **122**. This may, for example, be useful in some embodiments wherein a head pack may be re-useable and where it may be undesirable for water to enter the support layer **122**. In other embodiments, head packs may be designed for single use. In some of those embodiments, it may not matter that moisture vapor passes into the support layer **122** or transfer of moisture vapor to the support layer may be beneficial. Thus, the MVTR layer **120** may maintain a more constant air pressure than might otherwise be achieved if venting was done more often. This may be particularly beneficial in some embodiments described herein wherein the moisture vapor removal (MVR) layer itself is configured to help support a patient weight.

(37) In some embodiments, such as may be seen in FIG. **6**, the support layer **122** may comprise or consist of an inflatable chamber or bladder. The air and moisture vapor may be conveyed to the support layer **122**. For example, in the embodiment shown in FIG. **6**, the air and moisture vapor may be conveyed from the MVTR layer **120** to the support layer **122** through a fluid conduit **144**. A pressure relief valve **146** may be provided at an outlet **148** of the support layer **122** to hold the air until a threshold pressure is reached. When pressure in the support layer **122** reaches the pressure relief threshold, then the pressure relief valve may release the air and moisture vapor to atmosphere. Thus, air forced by the air mover (e.g., a pump as particularly shown in FIG. **6**) may both remove moisture vapor from the MVTR layer **120** and pressurize the support layer.

(38) In some embodiments, a flow rate of the pressure relief valve **146** (e.g., a flow rate supported when the valve is open) may be about equal or lesser than that of a flow provided by the air mover **116** (as configured in the system). For example, depending on the resistance to flow of the spacer layer **132** and fluid conduit **144** between the MVTR layer **120** and support layer **122**, an air mover or pump **116** may be configured to provide air flow to the support layer **122** at a certain rate. By using a foam in the support layer **122** with light compression force, the patients head will then be supported by the air at the desired pressure.

(39) In the embodiment shown in FIG. **6**, air flow is directed towards the support layer **122** using a fluid conduit **144**. As depicted, the fluid conduit **144** is shown as external tubing (outside of the support layer **122**). However, this need not be the case in other embodiments. For example, the fluid conduit **144** may be disposed within the support layer **122**. Alternatively, air flow may be provided through one or more holes formed in the boundary layer **134** of the MVTR layer **120**. For example, holes may be formed in the boundary layer **134** of the MVTR layer **120** so that moisture may be vented from the spacer **132**. For example, FIG. **12** shows an embodiment of a face pack **132** where moisture is removed from the MVTR layer **120** through one or more openings or holes

300 provided therein. The MVTR layer **120** is disposed on top of the support layer **122**. The boundary layer **134** of the MVTR layer **120** is generally disposed at the interface between the MVTR layer **120** and the support layer **122**. However, one or more holes or openings **300** are formed in the boundary layer **134**. The holes **300** may allow air to pass to the underlying support layer **122** for venting. Air may exit from the support layer **122** in any suitable way. For example, in the embodiment shown in FIG. **12**, air may exit through seams **304** formed in the support layer **122**. A plastic retaining shield **302** may be disposed underneath the support layer **122**. Straps **306** may be used to couple the face pack **132** to a head pack support assembly.

(40) The facial interface layer **130** may comprise one or more materials or layers which, individually or in combination, are liquid impermeable but permeable to liquid vapor. For example, the facial interface layer **130** may be comprised of a bonded material or laminate that includes any suitable "skin friendly" fabric on one side and an opposite side that is selectively permeable. A "laminate" as used herein may comprise a material formed with a plurality of layers. Or, the facial interface layer may include two distinct materials that are not bonded together, including a first material that is "skin friendly" fabric and a second material that is selectively permeable. The facial interface layer **130** may comprise a top layer that comprises a material suitable for skin contact, such as a material with low interfacial friction, high elasticity, and suitable hypo-allergenic properties. For example, in some embodiments, a "skin friendly" material or layer may be cotton or other suitable natural cellulose fiber and the second material or layer may be a suitable thermoplastic elastomer, or suitable fluoropolymer, including, by way of nonlimiting example, thermoplastic elastomer, PTFE (polytetrafluoroethylene), or a hydrophilic, polyether-ester block copolymer. The facial interface layer may be sealed along its perimeter to the boundary layer which creates an air-scaled chamber that may be filled by spacer layer.

(41) In some embodiments, the facial interface layer **130** may be made of a suitable moisture vapor permeable, reduced air permeability material. While it may be preferred for the air permeable layer to be fully air impermeable, it could be replaced with any fabric which limits air permeability through the layer. In some of those embodiments, air flow may be increased to offset the effects of the permeability.

(42) The spacer layer **132** may generally comprise any material which allows air to pass through or around its support structure. In some embodiments, a spacer layer **132** may comprise a support material that defines a scaffolding or matrix that allows air to pass in open spaces or interstices defined by the scaffolding or matrix. For example, in some embodiments, the spacer layer **132** may comprise reticulated foam, such as a polyurethane foam, which may be durable and porous.

(43) An air mover may be coupled to the spacer layer **132** through the air hose **118**. The air mover may provide positive pressure to force air into the spacer layer **132**, or negative pressure to draw air through the spacer layer. In some embodiments, the air mover may comprise a pump or a fan. In some embodiments, a positive or negative displacement air mover could be built directly into the face pack. Air provided by the air mover travels or circulates through the spacer layer, moving adjacent to the facial interface layer.

(44) As the patient perspires, the humidity at the patient/support interface increases. As the humidity increases at the interface layer, moisture is forced away from the patient's face, through the facial interface layer in the form of vapor into the drier air within the spacer layer. This action may continue as long as the air within the spacer layer is dryer than that at the patient interface. In some embodiments, air within the spacer layer may generally have a humidity that is about the same as the ambient environment. Ambient humidity is ideal for maintaining skin integrity. Thus, as long as equilibrium of humidity has not been reached between the spacer and facial interface layer, moisture vapor will be pulled from the facial interface layer to the spacer layer. In other embodiments, air within the spacer layer may be conditioned to a humidity level lower than that of the ambient environment. For example, if the kinetics of flow of vapor through the facial interface layer are slow, the facial interface layer and spacer layer may advantageously be operated under

nonequilibrium conditions. To help remove humidity an adsorption desiccant or dryer may be used, for example.
(45) The boundary layer **134** may, in some embodiments, comprise an air impermeable heat transfer vinyl. In such embodiments, heat transfer vinyl may be effectively sealed together with thermoplastic elastomers so that the facial interface layer **130** and boundary layer **134** may be sealed together using heat and pressure, for example. In some embodiments, a tie layer may be applied to either or both of the facial interface layer and boundary layer. A tie layer may be used to promote welding of two materials that otherwise may be difficult to bond.
(46) The MVTR layer **120** may be constructed in a variety of ways. In some embodiments, the facial interface layer **130** may be bonded to the air impermeable boundary layer **134** using a welding process or adhesive process. As used herein the term “welding” encompasses techniques for bonding two or more materials together using direct or indirect application of heat or to a boundary interface between two or more materials. Notably, some highly moisture vapor permeable fabrics may not typically be easily RF or sonically weldable, so some sort of adhesive may be necessary to bond the two layers. A seal tape may be used to seal a vapor permeable fabric to an impermeable layer. However, seal tapes are generally delicate, and use of seal tapes may be time consuming and difficult. Thus, use of a heat transfer vinyl as the impermeable boundary **134** provides certain advantages over other materials.
(47) Thus, in some embodiments, such as may be seen in the embodiment of FIG. **7**, to join and seal the MVTR layer a heat press or iron can be used. The facial interface layer **130** can be placed on a press. The facial interface layer **130** may, for example, comprise a skin friendly first side **133**. For example, as described above, the first side **133** may be made of a first material that is a “skin friendly” fabric such as cotton or other suitable natural cellulose fiber. In some embodiments, the first side **133** may be made of a synthetic fiber such as spandex or polyester. A second side **137** may be made of a second material such as a hydrophilic polyether-ester block copolymer. The spacer layer **132** may then be disposed against the facial interface layer **130**. The impermeable boundary layer **134**, may be constructed from heat transfer vinyl and may include a heat sealable side **135**. The boundary layer **134** may be cut to a similar shape as the facial interface layer **130**, may then be placed in contact with the spacer layer. The spacer layer **132** is cut to a smaller footprint than the facial interface layer and the boundary layer, such that it fits within the profile of the facial interface layer **130** and boundary layer **134** leaving perimeter facial interface layer and boundary layer to be sealed together. The stack of layers may be placed in a heat press. Once heat and/or pressure is applied via a platen, the spacer layer **132** may be pneumatically sealed within the facial interface layer **130** and the boundary layer **134** such as may be used to create a sealed air flow path.
(48) FIG. **8** shows an embodiment of a method **150** for making an MVTR layer **120**. As shown in step **152**, the one or more facial interface layer materials (e.g., a moisture permeable, air and liquid impermeable laminate) may be placed on a heat press or other suitable machinery used for welding. In step **154**, a spacer layer material may be placed over top of the one or more interface materials. The spacer layer material may be cut to a shape that defines a smaller footprint than the laminate. Accordingly, the spacer layer material may fit within the profile of the laminate. In step **156**, a boundary layer material (e.g., impermeable heat transfer vinyl) may be positioned on to top of the spacer layer. The boundary layer may, for example, be cut to a suitable shape so that it may be placed over the spacer layer. In a step **158**, heat may be applied so as to seal the facial interface layer to the boundary layer. One or more other steps may then be used to integrate the panel into a finished face pack, such as attachment to a support layer.
(49) Support Layer
(50) The support layer **122** may comprise any suitable support structure that tends to conform and distribute the weight of the patient's head across a larger area, such as foam. In some embodiments, the support layer **122** may comprise one or more foam layers having various spring rates or initial

force deflection ratings. For example, a plurality of different foams of different density may be used to provide multiple spring rates or initial force deflection ratings. The use of materials with graduated initial force deflection may be used to provide improved contact of the patient's face with the facial support, distributing force so as to reduce localized areas of high pressure on the face, such as bony protuberances. For example, a first layer may be more resistant to deformation than a second layer. This structure may help the support layer **122** deform in a way to reduce localized areas of high pressure on the face. In some embodiments, a support layer **122** having graduated initial force deflections could be replaced with a homogeneous support layer of foam or any suitable flexible material. The support layer **122** may be covered or enclosed by any suitable fabric.
(51) In other embodiments, the support layer **122** may include a “reforming” foam. For example, in the embodiment of FIG. **9**, reforming foam **160** is encased in a sealed cover **162**, which has a one-way check valve **164** (such as a ball check valve or a diaphragm valve) connected thereto and disposed so as to allow air to enter the support layer **122** when the check valve **164** opens. The one-way check valve **164** may be configured so that when a vacuum is created in the sealed cover (such as may be created when no support force is placed on the face pack, or when a support force is lifted from the face pack) air may be drawn into the support layer through the check valve. The support layer may include a pressure relief valve **146** at an outlet **148** of the support layer which can be selected or set to a specific relief pressure. As described in the previous embodiment, by using a foam with light compression force, the patients head will then be supported by the air at the prescribed pressure.
Further Embodiments
(52) In some embodiments, a patient may be supported with approximately uniform pressure across the surface of the face. For example, in the embodiment shown in FIG. **6**, the pressure provided across the surface of the patient's face may be approximately uniform. The pressure may be set and/or user selected within a pressure range. For example, the upper pressure limit may be the head pressure of the pump and the lower may be based on the reforming force of the encased foam. The supplied pressure of the embodiment may help to ensure that the pressure within the sealed support cover is always at the set check valve pressure, with the exception of small transient fluctuations.
(53) In some embodiments, a face pack may comprise of only the MVTR layer **120** (i.e., no support layer). In some of those embodiments, the boundary layer **134** may prevent air and moisture vapor from escaping, with moisture-saturated air being directed away from the patient and expelled from the MVTR layer **120** using a vent **143** or other suitable expulsion means. The MVTR layer **120** may, for example, be configured to both support the head of the patient, collect moisture vapor and route moisture vapor to the vent **143**.
(54) In some embodiments, a support structure and air pocket could be eliminated and only an air bladder, with an MVTR interface layer and similar mechanism of action could be used. In some embodiments, the spacer layer and boundary layer could be eliminated, and air could be pushed directly into the support layer. In other words, the MVTR layer and support layer could be combined into a single layer that provides both MVTR and head support. In some embodiments, the support layer may comprise only an inflatable chamber without any internal foam or other support structure. In such embodiments, the support layer, only pressurized air inflating the chamber may provide support.
(55) In some embodiments, a pressure relief valve **146** could be replaced with an orifice sized or adjusted to maintain a desired bladder pressure. For example, in the embodiment shown in FIG. **6**, pressure relief valve **146** may be replaced with or used in combination with an orifice. The orifice may be sized to work with the air pump **116** so that a desired pressure is maintained within the support layer **122**. In some embodiments, a pressure relief valve **146** may comprise an active electronically controlled pressure sensor and relief valve. This may, for example, be used to adjust a pressure setting so as to optimize support for individual patients. Still in a further variation of this embodiment, a valve **172** (such as an electromechanically controlled valve as further described in

relation to FIG. **10**) may be configured to operate in a pulsed, cyclical, pseudo-random, or other manner so that pressure within the support layer **122** may be varied over time. For example, in the embodiment shown in FIG. **6**, a valve **172** may be substituted for the relief valve **146**, and the valve **172** may be configured to periodically vent. For example, the valve **172** may operate under the control of a timing circuit so that pressure is periodically released.

(56) Periodic controlled venting or pressure cycling may help to provide effective removal of moisture vapor from the spacer layer **122** while helping to ensure patient comfort. Venting may, for example, be characterized by a frequency of venting and a time in which the valve may be open. During periods between venting, the air pressure in the support layer **122** and spacer layer **132** may stabilize to a relatively constant level set to an ideal range for comfortably supporting the patient. For example, pressure in the layers **122**, **132** may help to distribute force so as to reduce localized areas of high pressure on the face. During venting, moisture vapor may be removed so as to lower the vapor pressure of water in the spacer layer **132** thereby maintaining a required gradient of water partial pressures on either side of the interface layer **130** to ensure moisture removal. In some embodiments, the air mover or an inlet valve may be periodically cycled to stop air flow through the MVTR layer. The MVTR layer may thus deflate so that the head is supported more by the support layer, thus changing the support pressure points on the face. For example, pressure across the face may be more broadly distributed by the MVTR layer, but focused more on the forehead and bony protuberances of the face by the support layer. By alternating between supporting the face by the MVTR layer and supporting the face by the support layer, pressure points across the face may be changed periodically. Even with no air flow through the MVTR layer, drier air remaining in the MVTR layer will still tend to draw moisture vapor through the facial interface. After a suitable duration, such as a minute or two, the air mover or inlet valve may be cycled to force air through the MVTR layer, thereby re-inflating the MVTR layer and shifting the facial pressure points.

(57) In some embodiments, venting may be selected or controlled to ensure that a proper gradient of moisture is maintained across the facial interface layer **130**. For example, in some embodiments, air within the spacer layer **132** may generally be maintained so that it maintains a humidity that is about the same as the ambient environment. This may, for example, be controlled by selecting an optimum frequency of venting. However, pressure cycling may also be controlled so as to minimize disruption of pressure during venting. For example, in some embodiments, during venting, the valve **172** may only be opened for a period of time suitable so that the pressure in one or more of the layers **122**, **132** does not drop below a certain pressure. That is, the time of venting may be controlled so as to minimize variation of pressure in one or more of the layers **122**, **132** to help maintain proper support of the patient's head. For example, in some embodiments, the valve **172** may release pressure once within a time period of about 1 minute to about 60 minutes, or at some other suitable interval to balance between changing pressure and suitable moisture vapor venting. Thus, the valve **172** may be used in place of the valve **146** or used in combination with the valve **146** to provide for pressure cycling. Thus, the embodiment shown in FIG. **6**, may, for example, be used to provide for controlled pressure cycling similarly to the embodiments described below in relation to FIG. **10**.

(58) In some embodiments, multiple support bladders could be linked in series or parallel and multiple pressure relief and check valves could be incorporated to allow for different pressures in the bladders. Multiple pressure relief valves may be used to achieve proper flow and pressure in such a structure including multiple support bladders. For example, a first group of bladders may be positioned so that they significantly respond to venting when a first relief valve is activated. A second group of bladders may be positioned so that they significantly respond to venting when a second relief valve is activated. For example, in the embodiment shown in FIG. **6**, multiple support bladders could be arranged so that different pressures may be maintained across different regions of the head pack shown therein. Bladders in different regions of the support structure **122** could be filled or vented by coordinated control of multiple pressure relief and check valves so as to provide

a desired pressure to a head pack or to provide for pressure cycling. In some embodiments, multiple bladders of a support structure **122** may be arranged so that they respond at different rates to pressure changes (e.g., applied pressure from the air pump **116** or vented pressure initiated using the valves **146**, **172**). This arrangement could be used with multiple valves disposed at different locations or even using a single venting port. For example, pulsed venting using a pulsating valve **172** could initiate pressure variation in the different bladders based on different rates of fluid communication between the different bladders and the valve **172**. Thus, a pressure within a support layer **122** could be varied across different regions of a head pack using various combinations of one or more valves **146**, **172**. Thus, the embodiment shown in FIG. **6**, may, for example, be used to provide for controlled pressure cycling over different regions of a head pack as similarly described for the embodiment shown in FIG. **11**.

(59) In some embodiments, air pressure provided to one or more support bladders may be controlled by providing air pressure or relieving pressure (e.g., using one or more valves) in a time dependent manner or pulsed manner. For example, as shown in FIG. **10**, a face pack **170** may include one or more valves designed to selectively open or close so as to vary the pressure in the MVTR layer **120**. In this embodiment, the MVTR layer **120** may maintain sufficient pressure to help support the patient's face. For example, the MVTR layer **120** may comprise one or more bladders that work with the support layer **122** to help to support the patient. Alternatively, the MVTR layer **120** and support layer **122** could be combined into a single layer that provides both MVTR and full head support. As shown in FIG. **10**, the MVTR layer **120** may be connected to a valve **172**. The valve **172** may, for example, be disposed at an outlet region of the MVTR layer **120** positioned at a distal end away from the air pump **116**. Alternatively, more than one valve **172** may be used, including, for example, wherein selective venting of different bladders among a plurality of bladders is used.

(60) In one mode of operation, the face pack **170** may be used for removal of moisture vapor **142** while maintaining a minimal pressure in the MVTR layer **120**. For example, the valve **172** could be maintained in a closed position so that the pressure of the face pack **170** remains about constant after the MVTR layer is filled. Alternatively, the valve **172** may be operated in a mode wherein the valve periodically opens so as to temporarily relieve pressure from the MVTR layer **120** and/or allow for removal of moisture as described above. For example, the valve **172** may operate under the control of a timing circuit so that pressure applied to the patient's face is periodically reduced and water vapor removed. Venting from the valve **172** may, for example, provide for an about uniform decrease in pressure across the face pack **170**. Pressure within the MVTR layer **120** may then increase as air is provided from the air pump **116**, so that the pressure varies in a cyclic manner.

(61) As shown in FIG. **11**, a face pack **180** may include an MVTR layer **120** including a facial interface layer **130** and a boundary layer **134**. The MVTR layer **120** may receive fluid from air mover **116** (e.g., an air pump) through an inlet **200** connected to the fluid conduit **202**. A valve **184** may control flow of fluid from the air mover **116** to the MVTR layer **120**. The face pack **180** may further include a pressure support layer **186**. Pressure support layer **186** may, for example, comprise a bladder **188** with multiple pneumatically connected chambers. Pneumatic connections **204** between the chambers allow air to pass therebetween. Pressurization of the connected chambers may be used to create spatial variation of pressure within the support layer **186**. The pressure can be varied over time to such as by providing pneumatic pulses to the bladder **188** so as to offload pressure from different regions of the patient's face. In some embodiments, more than one bladder **188** may be used. With multiple bladders, the support layer **186** may provide different regions that are alternatively pressured. The pressure support layer **186** may be in fluid communication with the air pump **116** through an inlet **198** connected to the flow conduit **194**. A valve **182** may control flow from the air pump **116** to the pressure support layer **186**.

(62) In some embodiments, the face packs herein have a distinct advantage over fully pneumatic

bladders (no reforming foam) in that if pneumatic pressure is lost, the patient's head may still be supported by the support layer **122** (e.g., by foam inside the sealed support cover). For example, each of the embodiments shown in FIGS. **6**, **9**, **10**, and **11** may include a support layer **122** to support the patient even in the absence of air pressure.
(63) In further embodiments, a head pack as disclosed herein may be used for manual prone therapy. A patient turned face-down on a plain bed or other non-rotating patient support surface may be supported at the face by an MVTR face pack as disclosed herein. Thus, the disclosed face packs may be used for automated and manual proning. In some embodiments, packs as described herein could be used for a patient lying on their side. In those embodiments, cut outs may be provided for the patient's ears so that the patient is supported by other areas of the side of the patient's head.
(64) It is an objective of some embodiments herein to provide a head pack that protects a patient's skin from breakdown. The head pack may, for example, include means to help control the moisture surrounding the patient at a controlled level, means to help reduce shear force on the patient's skin or both. In some embodiments, a head pack may comprise a foam support of graduated initial force deflection, an active moisture vapor removal system, or a combination of both. Use of a support of graduated initial force deflection may help to reduce pressure on the face and reduce skin breakdown. Use of an active moisture removal system may help to control friction between the skin and the surface of the head pack so as to help reduce skin breakdown.
(65) It is an objective of some embodiments herein to provide a head pack that protects a patient's skin from breakdown by positively impacting various aspects of face packs and surrounding skin, including, for example, skin interface pressure, moisture, shear force, temperature, and combinations thereof. For example, interface pressure may be positively impacted by maximizing a surface area of contact area with the face pack while still allowing for visibility of the patient's eyes and maintaining adequate room for oral intubation lines. The pressure may be further managed through use of graduated density foam layers, which may allow for increased immersion and greater contact area.
(66) It is an objective of some embodiments herein to provide a method for making a face pack. For example, in some embodiments, an interfacial layer for a head pack may be bonded to an adjacent boundary layer using a welding process such as may exclude use of an external adhesive or external sealing tape. For example, in some embodiments, a heat transfer vinyl may be used to seal two or more layers of a head pack together.
(67) It is an objective of some embodiments herein to provide a support system for holding a patient face when suspended in a prone position or otherwise when rotating a patient over some angular range.
(68) Although the foregoing specific details describe various embodiments, persons of ordinary skill in the art will recognize that various changes may be made in the details of the disclosed subject matter without departing from the spirit and scope of the invention as defined in the appended claims and other claims that may be drawn to this invention and considering the doctrine of equivalents. Among other things, any feature described for one embodiment may be used in any other embodiment, and any feature described herein may be used independently or in combination with other features. Also, unless the context indicates otherwise, it should be understood that when a component is described herein as being mounted or connected to another component, such mounting or connection may be direct with no intermediate components or indirect with one or more intermediate components. Therefore, it should be understood that the disclosed subject matter is not to be limited to the specific details shown and described herein.

Claims

1. A proning face pack comprising: a moisture vapor removal layer configured for maximizing interface with the forehead and checks of a patient face, the moisture vapor removal layer configured to transmit moisture away from the patient face in the vapor phase while being impermeable to air and liquid phase water, the moisture vapor removal layer further shaped to conform to the outer dimensions of a patient's face, the moisture vapor removal layer further shaped to contour the patient's forehead and cheeks so as to form one or more openings for the eyes, mouth and nose of a patient's face; an air mover pneumatically coupled to the moisture vapor removal layer to move air through the moisture vapor removal layer; a support layer disposed adjacent the moisture vapor removal layer.
2. The head pack of claim 1 further comprising an air mover pneumatically coupled to the moisture vapor removal layer to push air into the moisture vapor removal layer; the support layer being pneumatically coupled to the moisture vapor removal layer so as to receive air from the moisture vapor removal layer, the support layer further comprising a vent disposed so as to pass air from the support layer to atmosphere.
3. The head pack of claim 2, the support layer further comprising a pressure relief valve coupled to the vent, the pressure relief valve configured to open at a set pressure point.
4. The head pack of claim 3, the pressure relief valve configured to permit air flow at a rate approximately equal to or lesser than a flow rate supplied by the air mover.
5. The head pack of claim 1, the moisture vapor removal layer being sealed along its perimeter to the support layer, the moisture vapor removal layer being separated from the support layer by an air impermeable boundary layer.
6. The head pack of claim 1, the support layer including a reforming foam, the support layer comprising a valve disposed so as to allow air flow into the support layer from atmosphere, the valve configured to open when the support layer is not supporting a patient face.
7. The head pack of claim 6, the moisture vapor removal layer comprising a vent disposed so as to vent moisture vapor to atmosphere.
8. The head pack of claim 6, the support layer further comprising a pressure relief valve configured to release air from the support layer at a predetermined pressure set point.
9. A proning face pack comprising: a moisture vapor removal layer configured for maximizing interface with the forehead and cheeks of the face patient's face, the moisture vapor removal layer configured to transmit moisture away from the patient's face as vapor while being impermeable to air and liquid phase water, the moisture vapor removal layer being shaped to conform to the outer dimensions of a patient's face, the moisture vapor removal layer further shaped to contour the patient's forehead and cheeks so as to form one or more openings for the eyes, mouth and nose of a patient's face and further configured to receive fluid pressure from a fluid pressure source so as to provide support to the patient when using the face pack; a vent disposed so as to vent moisture vapor from the moisture vapor removal layer to atmosphere; and a valve configured to pulse open so as to provide cyclical variation of pressure within the moisture vapor removal layer.
10. The head pack of claim 9 wherein said cyclical variation of pressure includes opening the valve at a frequency or for a period of time so as to maintain a suitable level of moisture vapor within the MVTR layer to encourage removal of moisture from the patient.
11. The head pack of claim 9 wherein said cyclical variation of pressure includes opening the valve at a frequency or for a period of time so that an air pressure in the moisture vapor removal layer does not decrease below a threshold level.
12. The head pack of claim 9 further comprising a support layer, including a material to help support the patient even if pressure is lost from the head pack.
13. The head pack of claim 9 further comprising a support layer, one or more openings made in the moisture vapor removal layer so as to allow moisture vapor to enter into said support layer.
14. The head pack of claim 9, said moisture vapor removal layer comprising: a facial interface

layer configured to interface with the patient's face, the facial interface layer being liquid vapor permeable but liquid and air impermeable; a boundary layer being air impermeable; and a spacer layer disposed between the interface layer and the boundary layer, the spacer layer configured to allow air flow therethrough.